Anirudh Yadav Boosa

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Summary

Undergraduate researcher in human-computer interaction (HCI) and accessibility, interested in human-AI interaction. B.S. in Computer Science expected May 2027; applying to PhD programs for Fall 2027. Co-author of a manuscript under review on Android apps for blind and low-vision users; now piloting how generative AI triages accessibility bug reports.

Education

University of North Texas Expected May 2027
B.S. in Computer Science *Denton, TX*
Sole-author senior thesis on attacks and defenses for drone control software, advised by Dr. Kirill Morozov (in progress).
Team capstone (in progress): IoT security monitor with screen-reader and audio alerts for blind and low-vision users.

Manuscript Under Review and Poster

R. Ghoneim, W. Aljedaani, **A. Y. Boosa**, and S. Ludi. "Evaluation of Blind and Visual Impaired Android Applications." *Universal Access in the Information Society* (Springer). Under review, 2026.

R. Ghoneim, R. Sharma, A. A. Islam, S. V. Pentyala, **A. Y. Boosa**, and A.-R. Glee. "What is Accessibility in Computing?" Poster, CRA UR2PhD program, 2025. Accepted to the Grace Hopper Celebration 2026 (canceled by its organizers).

Research Experience

Undergraduate Research Assistant, DiscoverABILITY Lab May 2025 – Jan 2026
University of North Texas. PI: Dr. Stephanie Ludi; day-to-day mentor: Dr. Rana Ghoneim *Denton, TX*

- Co-authored an empirical study of 26 Android apps built for blind and visually impaired users, pairing automated accessibility evaluation with 8,286 Google Play reviews of those apps (manuscript above).
- Ran Google Accessibility Scanner on onboarding, interactive and feedback screens and categorized 1,810 violations into 12 dimensions (touch target, contrast, metadata, focus order and others).
- Built Python pipelines (pandas, scikit-learn, NLTK) that classify the reviews into multi-label accessibility failure categories, and related scanner results to what reviewers reported to derive design recommendations for TalkBack users.

Research Projects

Open source. Code and documentation written with AI assistance, as each README states.

TalkBack Triage Pilot: exploratory interactive prototype for AI triage of accessibility bug reports Nov 2026
github.com/anirudh763/talkback-triage-pilot | live app: anirudh763.github.io/talkback-triage-pilot | JavaScript, Claude

- Proposed to Dr. Wajdi Aljedaani as a follow-up to the lab study: do TalkBack reproduction steps in a bug report change how a large language model (LLM) triages it? Built a web app that tests this on 75 real Android accessibility issues.
- Exploratory result, one LLM run: of 32 reports with TalkBack steps, removing only the steps switched the model to "needs more evidence" on 12 and the other way on 1 (exact McNemar $p = 0.003$). Manipulation check: the model named the steps as missing on 25 of the 32 once removed, against 1 of 32 when present.
- No human labels yet. The app lets raters label the same reports, and the labeling protocol (independent raters, shared rubric, agreement by Cohen's kappa) is written and public.

a11y-issue-warehouse: open dataset of accessibility bug reports (Python, SQL) Nov 2026

- 1,187 accessibility bug reports and a date-matched comparison sample (2,643 issues in all) from 15 open-source Android apps, in a tested SQLite warehouse with 19 data-quality checks; one check caught 180 keyword-search results that were not about accessibility. Labels are not hand-verified.

Skills

HCI and accessibility: accessibility evaluation (Google Accessibility Scanner, TalkBack, WCAG), qualitative and quantitative analysis of user feedback (app reviews, bug reports), scientific writing
Data, machine learning and programming: Python (pandas, NumPy, scikit-learn, NLTK, Matplotlib), text classification, SQL, generative AI (LLM) prompting and evaluation, C++, C, Java, JavaScript, HTML/CSS, Git, LaTeX

Awards and Activities

CRA UR2PhD Research Scholar (funded summer research and training, $8,000), 2025 | STARS AI Scholar, 2026–27 | Member, AccessComputing | Front Desk Assistant, UNT Housing (May 2025 – Present) | Languages: English, Hindi, Telugu